VIGNESH SUGUNATHAN

EMBEDDED FIRMWARE | PRODUCT DEVELOPMENT

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with hands-on exposure to the electronics product cycle: board bring-up, firmware development, integration testing and field validation. Built STM32, ESP32 and PIC based systems, including a solar LoRa network with 96.4% packet delivery in outdoor trials and a hazard-detection robot that placed runner-up nationally. Skilled at isolating hardware–firmware faults with oscilloscope, logic analyzer and SWD tools. Aims to help a product team deliver reliable firmware, faster bring-up and validation, and a smooth prototype-to-release path.

TECHNICAL SKILLS

Embedded & Firmware: Embedded C/C++, ARM Cortex-M (STM32F103), ESP32/ESP8266, PIC16F877A, 8051, MSP430, Arduino; Bare-Metal & HAL, GPIO, ADC, PWM, Timers, ISR, DMA, RTC, EEPROM, Low-Power, FSM
Protocols & Wireless: UART, SPI, I^2C, LoRa (SX1278, 433 MHz), Wi-Fi, Bluetooth/BLE, GSM, GPS (NEO-6M), AES-128, Packet Framing & CRC, RSSI Multi-Hop Routing
Hardware & Validation: Board Bring-up & Testing, Sensor/Actuator Interfacing, Motor Control (H-Bridge, BLDC, Stepper), Power-Rail Verification, Fault Isolation, Oscilloscope, Logic Analyzer, DMM, Soldering
EDA, Tools & Platforms: KiCad, Proteus, STM32CubeIDE, MPLAB X, Keil, PlatformIO, ST-Link (SWD), Git, Linux, ROS 2, Gazebo, RViz, Python

PROFESSIONAL EXPERIENCE

Robotics Instructor *Sep 2026 – Present*
NxtWave Institute of Advanced Technologies, Chennai

- Develop **robotics and drone** projects on **ROS 2 / Ubuntu** in **Python**, simulating robots in **Gazebo** and visualizing state, sensor streams and transforms in **RViz**.
- Build **IoT and robotics** prototypes on **ESP32 and STM32**, writing and debugging Embedded C/C++ firmware with sensors, motor drivers and Wi-Fi/Bluetooth telemetry.
- Perform hardware testing and fault isolation across UART, I^2C and SPI buses, power rails, wiring and sensor/actuator interfaces.

AI Robotics Engineer *Dec 2025 – Aug 2026*
Centrado Tech Solutions Pvt. Ltd., Chennai

- **Firmware development:** Wrote Embedded C/C++ for **ESP32 / STM32 (Cortex-M)**: GPIO, ADC, PWM, timers, interrupts and UART/I^2C/SPI drivers for ultrasonic, IR, IMU and environmental sensors.
- **Control & sensor fusion:** Implemented real-time sensor fusion and closed-loop motor control (PWM, H-Bridge, BLDC, stepper) for autonomous navigation and obstacle avoidance.
- **Board bring-up & testing:** Power-rail verification, peripheral validation and board-level functional testing; isolated faults with multimeter, oscilloscope and logic analyzer.
- **Prototyping & design:** Built harnesses, CAD models and 3D-printed prototypes; supported schematic/PCB reviews, tuned low-latency firmware and delivered benchmarked demos with documentation.

TOP ENGINEERING PROJECTS

Decentralized Solar-Powered Multi-Node LoRa Mesh Emergency Network *Field-tested*
LoRa SX1278 (433 MHz) | ESP32 | Arduino Nano | ESP8266 | GSM | NEO-6M GPS | AES-128 | Solar Power Mgmt

- Architected an off-grid, multi-node network with **dynamic RSSI-based multi-hop routing**, GPS location reporting, GSM alerting and **AES-128 encrypted** payloads to keep telemetry alive during disasters and cellular blackouts.
- Achieved **≈96.4% packet delivery over 500 transmissions at up to 200 m** in outdoor field tests; engineered solar harvesting and power management for continuous unattended node operation.

Autonomous Surveillance & Hazard Detection Robot *2025 | **Runner-Up, SRM National Robotics Hackathon***
ESP32 | Arduino Uno | MQ-2 Gas | HC-SR04 | L298N H-Bridge | Voice & IR Control | Blynk | Bluetooth SPP

- Built a **4WD mechatronic platform** with ultrasonic obstacle avoidance, line tracking, voice/IR/Bluetooth control, PWM motor speed control, and real-time toxic-gas (MQ-2) alerts with live Blynk telemetry.

STM32 Real-Time Environmental Monitor & I^2C OLED Driver *STM32F103C8T6 (Cortex-M3 @ 72 MHz)*
Embedded C | STM32CubeIDE | ST-Link V2 SWD | FTDI UART | I^2C | DS1307 RTC | DHT11 | SSD1306 OLED

- Performed bare-metal bring-up (HSE → PLL 72 MHz clock tree, I^2C fast-mode peripheral, timers/interrupts) and wrote register-level drivers for OLED display, DHT11 and RTC.
- Validated bus behavior and sensor acquisition through SWD in-circuit debugging, logic-analyzer decoding and a non-blocking 115200-baud UART diagnostic stream.

Microwave Oven Automation System (Embedded Systems Internship, Emertxe) *Jan – Feb 2025*
PIC16F877A | Embedded C | MPLAB X (XC8) | PICSimLab | GPIO | ADC | Timers | Interrupts | EEPROM | Matrix Keypad

EDUCATION

B.E. – Electronics and Communication Engineering *2022 – 2026*
Agni College of Technology, Chennai (Anna University) **CGPA: 8.2/10**

ACHIEVEMENTS & PUBLICATIONS

- **Best Research Paper Award, ICRIDET 2024 (IEEE)** for *Real-time Oil Spill Detection Using Deep Learning and CNNs*; second IEEE paper on dent damage detection (VIT Vellore).
- **Runner-Up**, SRM National Robotics Hackathon (2025) | NPTEL Embedded Systems and Design (2024)